COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
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SEATWORK NO. 1 **Algorithm, Flowchart, Pseudocode**

1. PROCEDURE OUTPUT

What procedures did you perform?

ILO A 1.1

- **Create a function that will return whether variable A is greater than variable B**

C: > Users > TIPQC > Downloads > ILO A 1.1.cpp

```cpp
#include <iostream>

bool isGreater(int A, int B){
    return A > B;
}
int main(){
    int A, B;
    std::cout <<"Enter value A: ";
    std::cin >> A ;

    std::cout <<"Enter value B: ";
    std::cin >> B ;

    if (isGreater(A,B)){
        std::cout << "A is greater than B" << std::endl;
    }
    else
        std::cout << "A is not greater than B" << std::endl;

    return 0;
    }
```

Image 1-1. C++ Structure Code for Answer

Analysis

In the code, the user is asked to input two numbers for A and B, and the system is programmed to compare the two numbers, if A is greater than B.

```
Enter value A: 12
Enter value B: 5
A is greater than B

--------------------------------
Process exited after 2.137 seconds with return value 0
Press any key to continue . . .
```

Image 1-2. ILO A output observations and comments.

ILO A 1.2

- **Create a function that will take two Boolean values and display the result of all logical operations then return true if it was a success.**

```cpp
#include <iostream>

bool logicalOP(bool A, bool B) {
    std::cout << "\nAll Logical Operations:\n";
    std::cout << "\tA AND B = " << (A && B) << std::endl;
    std::cout << "\tA OR B  = " << (A || B) << std::endl;
    std::cout << "\tNOT A   = " << (!A) << std::endl;
    std::cout << "\tNOT B   = " << (!B) << std::endl;

    return true;
};

int main() {
    bool A, B;

    std::cout << "\t\t  (0 or 1)\nInput value for A: ";
    std::cin >> A;

    std::cout << "Input value for B: ";
    std::cin >> B;

    if (logicalOP(A, B)) {
        std::cout << "\nSuccess. ";
    }

    return 0;
}

```

Image 2-1. C++ Structure Code for Answer

Analysis

The user is once again asked to input values (0 or 1) to variable A and B. It is then used to perform all logical operations.

```
                  (0 or 1)
Input value for A: 1
Input value for B: 1

All Logical Operations:
        A AND B = 1
        A OR B  = 1
        NOT A   = 0
        NOT B   = 0

True
```

Image 2-2. ILO A output observations and comments.

ILO B

C: > Users > TIPQC > Downloads > ILO B.cpp

```cpp
#include <iostream>
class Triangle{
private:
    double totalAngle, angleA, angleB, angleC;

public:
    Triangle(double A, double B, double C);
    void setAngles(double A, double B, double C);
    const bool validateTriangle();
};


Triangle::Triangle(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
void Triangle::setAngles(double A, double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);
}

int main(){
//driver code
Triangle set1(40, 30, 110);
if(set1.validateTriangle()){
    std::cout << "The shape is a valid triangle.\n";
} else {
    std::cout << "The shape is NOT a valid triangle.\n";
}
    return 0;
}

```

Image 3-1. C++ Structure Code for Answer

```
The shape is a valid triangle.

--------------------------------
Process exited after 0.107 seconds with return value 0
Press any key to continue . . .
```

Image 3-2. ILO B output observations and comments.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

- Create a C++ program to swap the two numbers in different variables.

C: > Users > TIPQC > Downloads > ILO C.cpp

```cpp
#include <iostream>

int main() {
    int A, B, C;

    std::cout << "Enter value for A: ";
    std::cin >> A;
    std::cout << "Enter value for B: ";
    std::cin >> B;

    std::cout << "Values before swap: \n\tA: " << A << "\n\tB: " << B << std::endl ;
    C = B;
    B = A;
    A = C;
    std::cout << "Values after swap: \n\tA: " << A << "\n\tB: "  << B;
}

```

Image 4-1. C++ Structure Code for Answer

Analysis

The user is asked to input values for A and B, afterwards the values assigned to A and B are swapped by assigning C to the value of B, B the value of A, then A the value of C which is the value of B.

```
Enter value for A: 1
Enter value for B: 2
Values before swap:
        A: 1
        B: 2
Values after swap:
        A: 2
        B: 1
--------------------------------
Process exited after 1.611 seconds with return value 0
Press any key to continue . . .
```

Image 4-2. ILO B output observations and comments.

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